Next Steps in Your AI Journey

Continue from structured-data projects into data science, unstructured data, generative AI, explainability, and MLOps, with drift monitoring and feedback-driven maintenance.

Original study notes based on Gwendolyn Stripling and Michael Abel, First edition, 2023. Book pp. 289–300. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.

Observe production data → Evaluate against labels → Detect drift/failure → Retrain or revise → Validate deployment

Detailed Notes

Going deeper and working with unstructured data

Source: book pp. 289–293 / PDF p. 305

Next steps include statistics, experiment design, feature engineering, model selection, software practices, and domain-specific learning. Text, images, audio, and documents need representations, labeling, and evaluation suited to their structure.

Unstructured data can still be paired with structured metadata and business outcomes. Start with a concrete decision, ensure privacy/rights, build a small evaluation set, and avoid treating a new modality as automatically useful.

Generative AI and explainable AI

Source: book pp. 294–297 / PDF p. 310

Generative AI produces new content; it introduces output-quality, safety, provenance, and factuality concerns beyond conventional fixed-label prediction. Use separate evaluation for correctness, usefulness, and risk.

Explainability methods such as feature attribution provide evidence about model behavior, but correlated variables and method assumptions can complicate interpretation. Explainability does not make a model causal or fair. Use explanations alongside domain review and data checks.

MLOps, drift, and continuous training/evaluation

Source: book pp. 298–301 / PDF p. 314

MLOps manages repeatable data/model pipelines, versioning, deployment, monitoring, and retraining. **Data drift** changes feature distributions; **concept drift** changes the relationship between features and labels. For example, customer demand may shift while product attributes stay unchanged.

Monitor input quality and delayed outcome labels. Continuous evaluation compares new labeled samples with expected performance; continuous training retrains on a schedule or trigger. Do not retrain blindly on recent data: validate labels, distribution, held-out performance, and rollback options before replacing a model.

Revision Check

1. What data or concept drift is plausible in your use case?
2. Which labels arrive late?
3. What evidence should trigger retraining?
4. How can an explanation be useful without being causal?

Production ML is an ongoing measurement and maintenance process, not a completed notebook.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.